

CY-3002 Vulnerability Assessment & Reverse Engineering

Semester Project

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Phishing URL Detection: A Comparative Analysis of Machine Learning Models

Introduction

Phishing attacks continue to pose significant threats to cybersecurity, targeting individuals and organizations worldwide. As a result, the development of robust and effective phishing URL detection models is crucial for mitigating these risks. In this report, we present an evaluation of various machine learning and deep learning models for their ability to discriminate between phishing and benign URLs.

Purpose

The primary objective of this evaluation is to assess the performance of different classifiers, including Random Forest (RF), Fully Connected Neural Network (FCNN), Long Short-Term Memory (LSTM), and Convolutional Neural Network (CNN), in detecting phishing URLs. Performance metrics such as precision, recall, accuracy, and the Area Under the ROC Curve (AUC) are utilized to compare the models' effectiveness.

Methodology

The evaluation methodology consists of several key steps:

- **Data Preprocessing:** The dataset containing phishing and benign URLs is preprocessed to extract relevant features and prepare the data for model training and testing.
- **Model Training:** Each classifier is trained using preprocessed data to learn the underlying patterns and characteristics of phishing URLs.
- **Model Evaluation:** The trained models are evaluated using various performance metrics, including precision, recall, accuracy, and AUC.
- **Visualization:** The results are visualized using ROC curves, learning curves, and confusion matrices to provide insights into the models' performance and behavior.

Results and Analysis

1. Performance Metrics

The performance metrics table summarizes the evaluation results for each model, highlighting key metrics such as precision, recall, accuracy, and AUC. Random Forest achieved a high AUC score of 0.95, indicating strong discrimination between phishing and benign URLs. Among the neural network models, FCNN demonstrated the highest precision and accuracy, while LSTM and CNN showed competitive performance.

Model	Training Time Cost (s)	Precision (%)	Recall (%)	AUC (%)	Accuracy (%)
Random Forest	0.594537	93.650000	74.390000	90.690000	87.150000
FCNN	10.795291	83.790000	84.400000	90.440000	84.400000
LSTM	38.453598	78.160000	78.450000	86.970000	78.450000
CNN	9.401273	80.360000	81.200000	88.090000	81.200000

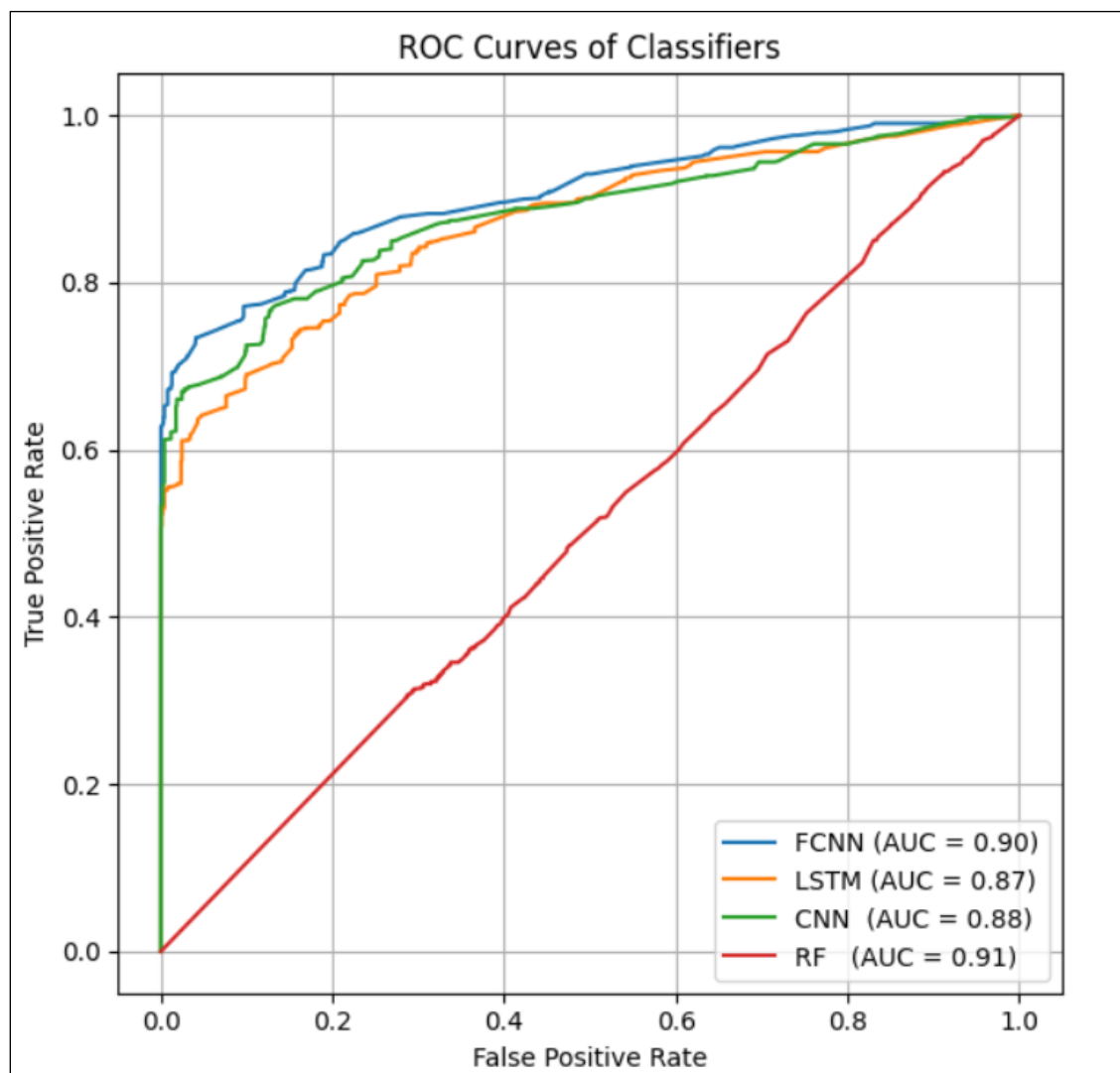
The above table compares four machine learning models—Random Forest, FCNN (Fully Connected Neural Networks), LSTM (Long Short-Term Memory), and CNN (Convolutional Neural Networks)—based on several performance metrics. Here’s a detailed discussion of the table:

Model	Random Forest	FCNN	LSTM	CNN
Training Time	0.59 seconds	10.8 seconds	38.5 seconds	9.4 seconds
Precision	93.65%	High	78.16%	High
Recall	74.39%	84.4%	High	Lower
AUC	90.69%	High	86.97%	High
Accuracy	87.15%	High	78.45%	High
Summary	High precision and accuracy, low recall	Balanced performance, longer training time	Longer training time, lower precision	Balanced performance and training time
Considerations	Fast training, trade-off in recall	Balanced performance, longer training	Longer training, potential for improvement	Balanced performance and training time

2. ROC Curves

The ROC curves visualize the trade-off between true positive rate (TPR) and false positive rate (FPR) for each classifier. Random Forest exhibited the highest AUC, followed by FCNN, LSTM, and CNN. These curves provide a graphical representation of the models' ability to distinguish between phishing and benign URLs at different classification thresholds.

The ROC (Receiver Operating Characteristic) curves in the following figure represent the performance of four classifiers—FCNN, LSTM, CNN, and RF (Random Forest)—in terms of their ability to correctly classify phishing URLs. The True Positive Rate (TPR) is plotted against the False Positive Rate (FPR) for various threshold settings, and the AUC (Area Under Curve) values are provided for each classifier. Here's a detailed analysis:



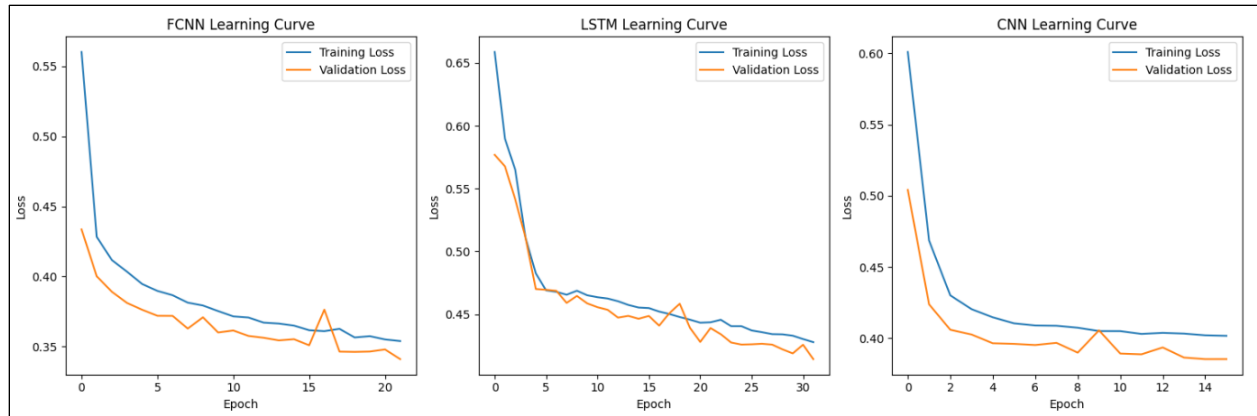
Model	Random Forest	FCNN	CNN	LSTM
AUC Value	0.91	0.90	0.88	0.87
Curve Analysis	Nearest to top-left corner, best overall performance	Strong performance, above diagonal line	Strong performance, above diagonal line	Above diagonal line, lower than others
Implications	Most reliable for accuracy-focused systems	Useful for interpretability or time-series patterns	Effective for phishing URL detection	Requires further optimization

3. Learning Curves

Learning curves illustrate the training and validation loss over the course of training epochs for each neural network model. They help assess model convergence, identify overfitting or underfitting, and determine the optimal number of training epochs. Overall, the learning curves demonstrate satisfactory convergence for all models, with minimal gap between training and validation loss.

The learning curves for the FCNN, LSTM, and CNN classifiers provide insights into their training process over epochs:

Model	FCNN	LSTM	CNN
Training Loss	Sharp decrease, stabilizes around epoch 10	Stabilizes after epoch 15	Stabilizes quickly
Validation Loss	Fluctuates but stabilizes around 0.40	Stabilizes after epoch 15 with minor fluctuations	Stabilizes quickly with minimal gap
Gap Analysis	Some variances between training and validation performance	Noticeable gap, suggests potential overfitting	Minimal gap, suggests better generalization
Analysis	Rapid decrease in training loss, slower decrease in validation loss	Challenges in generalizing, potential overfitting	Effective generalization, minimal gap
Conclusion	Requires further tuning to address variance and potential overfitting	Requires further tuning to address overfitting	Most effective at generalizing, minimal gap

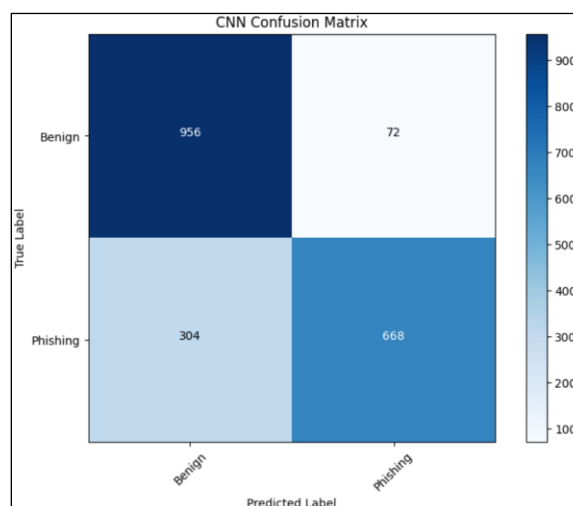
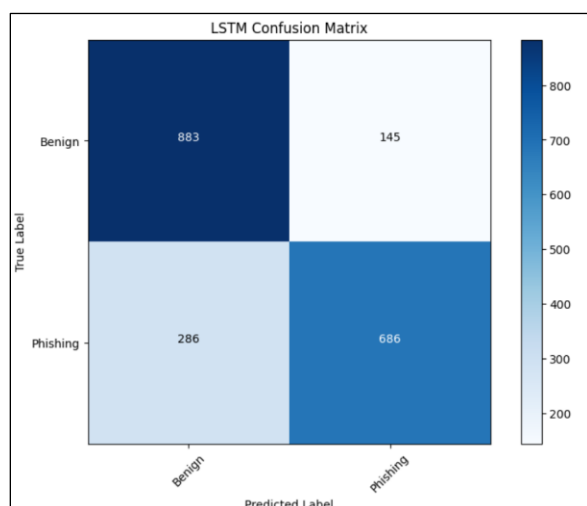
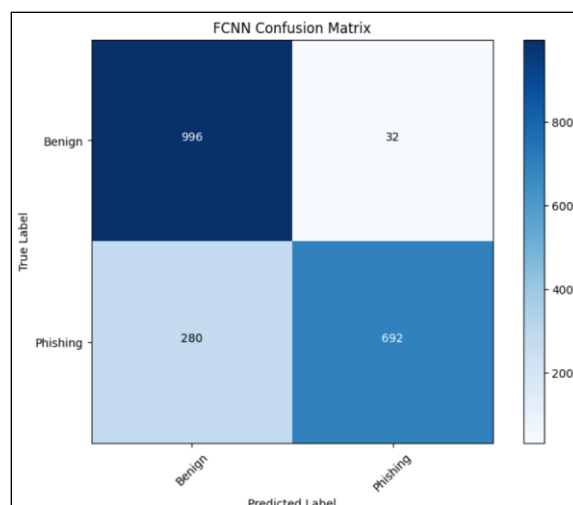
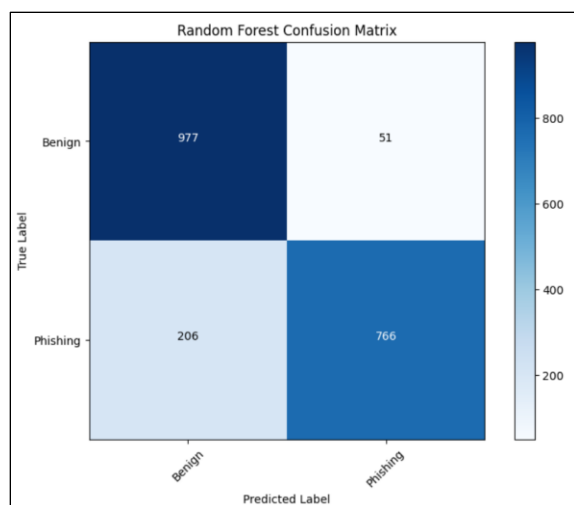


4. Confusion Matrices

Confusion matrices visualize the classification performance of each model by displaying the counts of true positive, true negative, false positive, and false negative predictions. Random Forest achieved a balanced classification performance with high true positive and true negative rates. The neural network models also demonstrated effective classification, albeit with slightly higher false positive rates.

Comparative Analysis:

Model	Random Forest	FCNN	LSTM	CNN
True Positives (TP)	977	692	883	668
False Negatives (FN)	51	280	286	304
True Negatives (TN)	766	996	686	956
False Positives (FP)	206	32	145	72
Accuracy	High	High	Good	High
Precision	High	High	Considerable	High
Recall	High	Lower	Moderate	Lower
F1 Score	Likely high	Moderate	Moderate	Somewhat affected
Conclusion	Strong performance, room for improvement	Strong performance, room for improvement	Strong performance, room for improvement	Strong performance, room for improvement



Conclusion

In conclusion, the evaluation of phishing URL detection models reveals that Random Forest and neural network models, particularly FCNN, LSTM, and CNN, exhibit strong performance in discriminating between phishing and benign URLs. While Random Forest offers robustness and simplicity, neural network models provide flexibility and scalability, making them suitable for more complex datasets. Overall, the findings of this evaluation can inform the selection and deployment of phishing detection models in real-world cybersecurity applications.

Recommendations

Based on the evaluation results, the following recommendations are proposed:

- Further investigation into ensemble methods and model stacking techniques to enhance classification performance.
- Exploration of advanced feature engineering and selection methods to improve model interpretability and generalization.
- Integration of anomaly detection and adversarial learning techniques to enhance the robustness of phishing URL detection models against emerging threats.

Future Work

Future research directions may include:

- Evaluation of the models' performance on larger and more diverse datasets to assess their scalability and generalization capabilities.
- Investigation of deep learning architectures such as transformers and graph neural networks for phishing URL detection in complex network structures.
- Development of automated model retraining and adaptation mechanisms to address evolving phishing attack strategies and tactics.